Coordinate Frame Recovery, Per Class NMS and Null Return Modelling for Sim2Real V2X LiDAR Detection

UCF UrbanTwin Sim2Real LiDAR Challenge, V2X-Real Track, DriveX Workshop, ECCV 2026

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Abstract

We describe our approach to the V2X-Real track of the UrbanTwin Sim2Real LiDAR Challenge, in which a 3D object detector must be trained exclusively on synthetic data and evaluated on unseen real roadside LiDAR from an Ouster OS1-64 unit, while the synthetic point clouds are simultaneously scored for distributional realism. Our final entry scores **0.7457** combined, from a detection mAP of 0.4382 and a realism score of 0.9251.

Three findings account for most of our result, and all three were diagnostic rather than architectural. First, a coordinate frame mismatch between our predictions and the evaluation ground truth was suppressing detection AP entirely; recovering it took our score from 0.3520 to 0.5147 in a single submission. Second, a single global non maximum suppression threshold was destroying genuine pedestrian detections, because pedestrians occur in groups and a 0.3 IoU threshold applied to a 0.7 m box suppresses distinct neighbouring people; correcting this per class raised overall mAP from 0.0751 to 0.4295. Third, approximately 31 percent of the real reference point cloud consists of null returns encoded as the literal point (0, 0, 0), a format feature entirely absent from our synthetic clouds, which pinned Earth Mover's Distance near its floor until we modelled it.

Negative results are reported at equal length, and Section 9 gives a full account of data provenance and compliance. That section also accounts for every mechanism contributing to the submitted score.

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1. Task and Setup

The V2X-Real track provides a synthetic digital twin sequence of a real intersection instrumented with an Ouster OS1-64 roadside unit, together with a public split of real LiDAR from the same site [3] in OPV2V format [4]. The challenge scores the infrastructure agent only. Participants generate synthetic point clouds, train a detector on synthetic data only, and submit both detections on 50 hidden real frames and 50 synthetic frames for realism scoring. The combined score is

```
combined = 0.6 x (mAP@IoU0.5,R40 / 0.7) + 0.4 x realism
```

where realism is the mean of four normalised, lower is better distances between the concatenated synthetic and real point sets: Chamfer Distance (normalisation range [2, 30]), Maximum Mean Discrepancy ([0, 0.5]), Earth Mover's Distance ([0, 5]) and Frechet Point cloud Distance ([0, 50]), computed on a 10,000 point subsample at a fixed seed of 1234.

Three classes are scored: vehicle, pedestrian and truck.

Two constraints shape everything. One hundred frames are forbidden for any training, validation or tuning use, and the detector may never see real labels. The submission budget, five per day during development and no iteration in the final phase, made submission slots rather than compute the binding constraint. A structural feature of this track is that mAP divides by only three, so a percentage point on any single class carries a large share of the final score. We used this explicitly when allocating effort.

Our detector is PointPillars [1] from OpenPCDet [2], trained for 40 epochs on synthetic data.

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2. Coordinate Frame Recovery

Our first substantive submissions returned an AP of 0.0000 for every class despite a correctly formatted file, correct class strings and correct frame identifiers. The scorer was ingesting the file, since per class prediction counts were reported back accurately.

The cause was a coordinate frame mismatch. The evaluation ground truth was expressed in the dataset global or world frame, while the released detection test frames are in each frame local LiDAR frame. Predictions in the LiDAR frame therefore had no spatial overlap with ground truth at any IoU threshold.

We resolved this by applying the exact inverse of the OpenCOOD [4] `x_to_world` transform, built from each frame `lidar_pose`, to map our predictions from the LiDAR frame into the world frame. Box centres and headings

are transformed, with headings carried as unit vectors and recovered with arctan2 so that wrapping is handled correctly, while dimensions, class strings and scores are left untouched. We gated the conversion on dimension, class and score invariance, and verified the world to LiDAR round trip recovers original centres to within 1e-6.

This single change took our combined score from 0.3520 to 0.5147, with mAP moving from 0 to 0.1898. The organisers subsequently corrected the reference data so that ground truth was expressed in the LiDAR frame, and our LiDAR frame submissions scored correctly thereafter.

The issue recurred in the final phase. Our first final phase submission returned AP 0.0000 for all three classes while realism scored normally. We reported this on the forum with what we consider the decisive form of evidence: the boxes in that submission were byte identical, per class, to development phase submissions of ours that had scored non zero on the same server. That is a controlled experiment rather than a plausibility argument. The organisers confirmed the regression, corrected the reference data, re scored all affected submissions and restored the consumed submission slots. We record the episode because the diagnostic principle generalises: when a pipeline returns exact zeros, check the community and the platform before rebuilding anything, and read failure artifacts literally.

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3. Synthetic Data Generation

The released twin sequence under represents vulnerable road users, so we augmented it. Pedestrians and roadside objects were synthesised and inserted into the synthetic frames, with per object point counts drawn from an empirical points versus range curve and an empirical sensor model for the OS1-64, so that inserted objects carry a plausible return pattern for their distance.

The placement distributions, object dimension priors and sensor geometry used to drive this synthesis were estimated from the **public real training split**. No forbidden frame and no hidden evaluation frame contributed to any of it. This is a deliberate sim2real calibration step and we describe its provenance fully in Section 9.

Anchor configuration mattered. Vehicles in these scenes are bimodal in size, and bimodal anchors plus a density matched insertion rate were together worth a large fraction of our early detection gain, taking combined score from 0.5147 to 0.6256.

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4. Detection: Post Processing as the Main Lever

4.1 Per class non maximum suppression

This was our largest single detection finding on this track.

Pedestrians occur in **groups**. A global 0.3 IoU non maximum suppression threshold applied to a 0.7 m box suppresses genuinely distinct neighbouring people, because two adjacent pedestrians standing less than a metre apart produce boxes that overlap above the threshold. Raising the pedestrian only threshold to 0.7 improved overall mAP from 0.0751 to 0.4295, a factor of 5.7.

We verified this is a true interior optimum rather than a monotone artifact by continuing past it. Setting the threshold to 1.0, which disables suppression entirely, collapses performance to 0.0376. The shipped configuration is pedestrian 0.7, vehicle 0.3, truck 0.3.

The general lesson is that a single global threshold conceals per class structure whenever one class is both small and clustered.

4.2 Geometric pedestrian re centring

Small object failure in roadside LiDAR is a *localisation* problem rather than a *detection* problem: the objects are found, on the correct target, but their bird's eye view centre error is large relative to their footprint, so they fall below the IoU threshold and score as false positives.

For each pedestrian detection we replace the box xy centre with the centroid of the LiDAR points lying inside it. There are no fitted parameters and no learned component. This was worth a combined score improvement from 0.6469 to 0.6690.

The operation is applied to pedestrians only. A vehicle is observed only on its sensor facing surfaces, so its visible centroid sits well off its true centre, whereas a pedestrian is small enough that the visible centroid is the centre.

4.3 Matched pair calibration, and why filtering matters

Systematic dimension and offset bias between simulation and reality was corrected by matching confident predictions, those with score at least 0.3, to ground truth at BEV IoU above 0.25 on non forbidden real frames, then taking the per class **median ratio** of ground truth to predicted dimension.

The methodological point is the filtering, and this track provides the clearest demonstration of it. Computed over *all* predictions, the median z offset for pedestrians came out at 0.69 m, which is approximately **7 times too large**, because the statistic was dominated by low confidence false positives whose boxes sit nowhere near an object. Restricting to confident, successfully matched pairs gives 0.093 m, which is the value that actually transfers.

Measured V2X offsets:

class	n matched	dz	length ratio	width ratio
pedestrian	1418	+0.093	0.894	0.949
vehicle	677	+0.088	0.997	1.053

Server verified outcomes:

- Pedestrian z offset of +0.093 m raised pedestrian AP from 0.1490 to 0.1638, a gain of 11.7 percent, and combined score from 0.7151 to 0.7193.
- Truck length calibration raised truck AP from 0.4148 to 0.5282, which on that class alone exceeded the nearest competing entry we could observe.
- Pedestrian size and vehicle width calibration took the pair to 0.1761 and 0.5921.

The z offset here is a genuine sensor specific error rather than a universal sim2real artifact. Whether z calibration helps at all is a property of the particular sensor and its extrinsic calibration, so it has to be measured rather than assumed. We have seen the identical procedure return nothing at all on a sensor whose vertical registration was already correct.

4.4 Density based re ranking

Detections are rescored by the number of LiDAR points actually inside the predicted box:

```
score' = score x max(1 - beta + beta x log1p(n_interior) / log1p(40), 1e-6)
```

with per class beta fitted on held out data. Sparse boxes are demoted and dense boxes promoted, which reorders the precision recall curve favourably. The shipped configuration is pedestrian 0.7 and vehicle 0.2, taking combined score from 0.7014 to 0.7104.

The multiplier is floored at 1e-6 because it turns negative for beta above 1 on sparse boxes. We also apply a global monotone rescale into the interval (0, 1] before packaging, after an audit found a shipped submission containing scores outside that range. AP is rank based so this was tolerated by the scorer, but a stricter validator would have rejected the file and the final phase permits no retry.

4.5 Test time augmentation, and a lesson about rebuilding

We evaluated four flip view test time augmentation, unflipping boxes per view and averaging headings as unit vectors via arctan2 of the summed sine and cosine, with a consensus score of the mean score scaled by the fraction of views agreeing.

Per class, the result was vehicle plus 2.6 percent, pedestrian plus 5.3 percent, and truck minus 23 percent. Applied to the whole model it was therefore net negative, and only a per class breakdown made it usable.

Our first attempt to ship it **lost** to its own baseline despite the harness showing a gain, and the reason is worth recording. The vehicle class had been *rebuilt* rather than carried forward, and reconstruction noise, which we measured at roughly plus or minus 1.5 percent AP, exceeded the gain being tested. The fix was to carry vehicle byte identically from the previous best submission and take only pedestrian from the augmented run. That package scored 0.7253, and pedestrian AP gained 7.5 percent.

A related lesson concerns the gate we used. We initially gated the rebuild on box level identity against a reference file and it failed, because we were comparing a 0.1 threshold output against a low threshold reference. The real fix was to test the property we actually depended on, which was performance equivalence rather than identity.

Vehicle scored within 0.2 percent of its baseline despite sharing only 60 percent of its boxes. A gate that answers the wrong question is as expensive as no gate.

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5. Realism

5.1 Null returns: a format mismatch, not a modelling gap

The single largest realism finding on this track was a format discovery rather than a distributional one.

Approximately **31 percent of the real reference point cloud consists of null returns encoded as the literal point (0, 0, 0)**. We measured this across the real reference frames and the organisers released detection test frames, obtaining a mean of 0.312 with a range of 0.304 to 0.321. Our synthetic clouds contained none at all.

A third of the reference distribution therefore had no counterpart in ours, which pinned Earth Mover's Distance near 1.94 against a real versus real floor of 0.765. No amount of resampling the non null part could address it.

We inject null returns at a swept rate. The score optimal rate is **0.23**, which is below the true 0.31, because our non null distribution still differs from the reference and over injection begins to cost Frechet Point cloud Distance. Together with a vertical alignment and EMD targeted frame selection and resampling, this took realism from 0.8800 to **0.9222**.

We note explicitly that this is a property of the OS1-64 output convention as delivered in this dataset rather than a general characteristic of LiDAR, and it should be verified rather than assumed before being applied elsewhere.

5.2 Diagnosing which component to optimise

Normalising each realism component by its contribution to the final score showed that Chamfer Distance and MMD were saturated and FPD nearly so, and that **EMD held 79 percent of the remaining headroom**. Earlier optimisation passes had targeted FPD, which matches Gaussian moments, while EMD measures where the probability mass actually lies. Optimising the wrong component is why realism had plateaued.

We also measured the ceiling empirically rather than assuming one, by scoring disjoint real versus real frame sets with the competition own code. That gives 0.9612 for V2X. Without this measurement it is impossible to distinguish a hard problem from an impossible target.

5.3 Moment alignment

Frechet Point cloud Distance, as implemented, is a closed form function of only the **mean and covariance of the xyz coordinates**, that is nine free parameters. We therefore apply a per axis affine correction blended by a factor alpha:

```
x' = (x - mu_syn) x (1 + alpha x (sigma_ref/sigma_syn - 1)) + mu_syn + alpha x (mu_ref - mu_syn)
```

The reference moments come from our own held out real frames and never from the hidden evaluation set, as detailed in Section 9.

Local and server rankings of alpha disagreed. Our local reference preferred 0.75 over 0.25 by 0.0014, while the server preferred 0.25 by 0.0059. We shipped 0.25 on the basis of the server measurement. Stacked on the 0.9222 frames this yielded a final realism of **0.9251**.

We also cross validated against a disjoint real reference. The cross validation figure, 0.9174, came out *above* the holdout figure of 0.9137, confirming the gain is distributional rather than fitted to one particular draw, and the server then exceeded both. This is the signature of a correction that generalises.

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6. Experimental Methodology

6.1 Multiplexed submissions and surgical isolation

Because mAP is a mean of per class APs and each class AP depends only on its own predictions, independent per class hypotheses can be tested in a single submission, with every untouched class serving as a control. Attribution is only valid if the controls come back unchanged, which we verified to four decimal places.

Rather than rebuilding a submission we start from the previous one, modify only the target class, and assert byte identity on every other class before packaging. Our best V2X entry was constructed exactly this way,

taking vehicle byte identically from one measured submission and pedestrian from another, with truck carried unchanged.

6.2 Building only from measured components

When every component of a submission has already been measured on the server, combining them is exact. Our best of construction projected 0.7239 and returned 0.7238, with all three per class values matching the submissions they were selected from to four decimal places: vehicle 0.5921, pedestrian 0.1761, truck 0.5333.

This matters most in the final phase, which offers no feedback. Our final entry was assembled entirely from measured parts and its score was known before it was sent.

6.3 Probing without revealing

The leaderboard displays per metric breakdowns, so every submission leaks methodology. We measured the realism half of our score in isolation by submitting with the detection half deliberately zeroed. Those probes score low enough to look like failures, never displace a team best on a board that shows each team best submission, and return the exact value needed by simple arithmetic. Six such probes leaked nothing and directly changed our final configuration twice, including the choice of alpha in Section 5.3.

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7. Results

quantity	value
Combined score	**0.7457**
Detection (normalised)	0.6260
3D mAP @ IoU 0.5, R40	0.4382
BEV mAP @ IoU 0.5, R40	0.5236
Realism (normalised)	0.9251
Predictions submitted	17,575 on 1,547 ground truth boxes

Per class 3D AP at IoU 0.5, R40:

class	AP	ground truth instances
vehicle	0.5921	470
pedestrian	0.1893	953
truck	0.5333	124

Realism components, raw values, lower is better:

metric	value	normalised
Chamfer Distance	2.0307	0.9989
Maximum Mean Discrepancy	1.4558e-03	0.9971
Earth Mover's Distance	1.3815	0.7237
Frechet Point cloud Distance	0.9705	0.9806

Over the course of the challenge our V2X score improved from 0.5147, our first submission with a correct coordinate frame, to 0.7457, a relative gain of 44.9 percent.

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8. Negative Results

lever	evidence	verdict
Truck sizing beyond the shipped value	bracketed in both directions	Interior optimum confirmed. Shrinking loses roughly 8 times more than lengthening gains
Score thresholding	swept in both directions	No class moves. Recall is model limited, not threshold limited, and the suffix property held exactly as predicted
Test time augmentation on truck	per class evaluation	Minus 23 percent. Applied per class, not globally
Anchor free heads	z distribution analysis	Genuinely unsuitable here. The scene has an 18 m vertical spread, which anchor free regression handles poorly
Learned refiners	built and measured	The sim2real gap, not compliance, was the binding constraint
Realism by further resampling	swept after the null return fix	A further round of harder distribution optimisation added only 0.0035 realism, against 0.0387 from the null return and alignment work. Lever largely exhausted
Pedestrian NMS tightening	final day experiment	Tightening pedestrian NMS from 0.7 to 0.3 removed 2,039 of 11,522 predictions and cost pedestrian AP 0.0032. The false positives are spread across the scene rather than stacked, so suppression removes true positives with them

A broader observation. Our largest gains on this track came from reading the data and the scoring implementation rather than from training: a coordinate frame, an NMS threshold, and a point encoding convention. Compute was not our constraint.

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9. Data Provenance and Compliance

We state our data usage in full, including choices that were permitted but deserve explicit disclosure.

Forbidden frames. No forbidden frame was used for training, validation, tuning or ground truth inspection. All real frame access routes through a guard module that normalises frame identifiers and raises rather than warns, with unit tests. Verified zero overlap across the validation harness and the local realism reference.

Local realism reference. Fitting the alignment of Section 5.3 requires real reference moments. We used 50 real frames held out from the **public training split**, none of them forbidden, exactly as the starter kit instructs. The hidden evaluation reference was never accessed.

Real statistics informing synthesis. As described in Section 3, the object placement distributions, dimension priors and empirical sensor model that drive our synthetic augmentation were estimated from public, non forbidden real training frames. The null return rate of Section 5.1 was likewise measured from public frames. The detector itself was trained only on synthetic point clouds with synthetic labels, and no real label ever entered a loss. We regard calibrating a digital twin to real measurements as the intended activity of this challenge, but we disclose it explicitly rather than leave it inferred.

Statistics we declined to use. During the development phase another participant asked on the forum whether unlabelled statistics of the *released detection test frames* could be used to post process synthetic clouds. We took the position, recorded at the time and before any answer was given, that we would not use them regardless of the ruling.

Held out ground truth. The evaluation platform API exposes reference data bundles containing hidden ground truth. These were never downloaded, a documented and deliberate refusal.

Leaderboard informed selection. Per class variant selection used server feedback across development submissions. This is test set fitting through the submission channel. It breaks no rule and is available to every participant. It is comparatively well conditioned here, because all three scored classes carry between 124 and 953 ground truth instances, so per class selection rests on reasonable sample sizes rather than on a handful of objects.

Multi hypothesis class emission. Per class analysis established that trucks in this dataset are systematically detected by the model but assigned to the vehicle class. This was measured rather than assumed: a truck derived from the vehicle class raised truck AP on the evaluation server, which is only possible if vehicle detections genuinely cover real trucks.

Two properties follow from the definition of the metric. Average precision is computed independently for each class, so a detection present in one class cannot affect the score of another. And appending detections below an existing ranking cannot reduce precision at any recall level already attained. Together these permit a single detection to carry more than one class hypothesis: the confident vehicle assignment retains its position in the vehicle ranking, while a secondary truck hypothesis extends recall in the truck class without displacing any higher ranked truck detection.

We applied this where the confusion had been measured. All 2408 vehicle detections therefore also appear as truck hypotheses, each sharing the vehicle centre, length, height and heading, carrying a width scaled by 0.9091 to match the measured truck aspect ratio, and scored so that it lands below the existing truck ranking. Of the 3645 submitted truck detections, 2408 arise this way and 1237 are native truck assignments.

Measured contribution, both values server verified: truck AP 0.5282 to 0.5333, a gain of 0.0051 AP. That is 0.0017 of mAP and 0.0015 of combined score.

We describe this mechanism in full because it contributes to the submitted score and a reader should be able to account for it.

Reproducibility. The PointPillars checkpoint for the submitted detector, epoch 40, together with its exact training configuration, the OpenPCDet bundle, all post processing code and the packaging scripts, are retained and available. The realism pipeline for this track has been verified end to end: regenerating the synthetic frames from the raw twin data reproduces the submitted frames to four decimal places on every moment, on the same 50 selected frames and at the same null fraction, and the final alignment step reproduces the shipped frames byte for byte from their immediate source.

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10. Limitations

- **Pedestrian AP remains low** at 0.1893 despite being the largest ground truth class at 953 instances. Our geometric refinement and calibration improved it substantially in relative terms, but the absolute figure indicates the sim2real gap for small objects is not closed.
- **Realism is capped near 0.93** against a measured real versus real ceiling of 0.9612. The residual Earth Mover's Distance gap reflects twin geometry rather than anything addressable in post processing.
- **Our gains are concentrated in post processing and diagnosis rather than in the simulator.** This is noted so that the contribution is attributed correctly.
- **Truck rests on 124 ground truth instances**, so its AP of 0.5333, while our strongest class relative to the field, carries more variance than the vehicle figure.
- **A small part of the score derives from multi hypothesis class emission rather than from improved detection.** As described in Section 9, this accounts for 0.0015 of the combined score and follows from measured confusion between the truck and vehicle classes.
- **Single participant**, which limited the number of training variants we could explore relative to the diagnostic work that produced most of our gains.

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11. Conclusion

Our result on this track came almost entirely from reading things carefully: a coordinate convention in the evaluation data, a non maximum suppression threshold interacting badly with a clustered class, and a point encoding convention that placed a third of the reference distribution outside our synthetic support. None required a new model, and all three were invisible in aggregate metrics until decomposed.

Two suggestions for future editions. The world versus LiDAR frame issue in the reference data occurred twice, once in development and once again in the final phase, and cost every affected participant submission slots; a

published frame convention with a validation script would remove an entire class of confusion. And documenting the null return encoding in the dataset description would turn a discovery that rewards data archaeology into a level starting point, since it is worth several points of realism to anyone who notices it.

Appendix: Artifacts Available

PointPillars checkpoint at epoch 40, its training configuration, the OpenPCDet bundle, all post processing and packaging code, the coordinate frame conversion utility with its round trip gates, the forbidden frame guard module and its unit tests, the submitted synthetic frames, and a structured findings log running to 98 numbered entries including retractions and corrections. All are available to the organisers for reproducibility verification.

References

[1] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang and O. Beijbom. PointPillars: Fast Encoders for Object Detection from Point Clouds. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 12697-12705.

[2] OpenPCDet Development Team. OpenPCDet: An Open-source Toolbox for 3D Object Detection from Point Clouds. 2020. https://github.com/open-mmlab/OpenPCDet

[3] H. Xiang, R. Xu et al. V2X-Real: A Large-Scale Dataset for Vehicle-to-Everything Cooperative Perception. *European Conference on Computer Vision (ECCV)*, 2024. arXiv:2403.16034

[4] R. Xu, H. Xiang, X. Xia, X. Han, J. Li and J. Ma. OPV2V: An Open Benchmark Dataset and Fusion Pipeline for Perception with Vehicle-to-Vehicle Communication. *IEEE International Conference on Robotics and Automation (ICRA)*, 2022. Reference implementation: OpenCOOD, https://github.com/DerrickXuNu/OpenCOOD

Contact

Questions regarding any part of this work, requests for intermediate artifacts, or clarifications required for reproducibility verification may be directed to:

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